

Kristine Stanley

Staff / Senior Rust Software Engineer

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PROFESSIONAL SUMMARY

Staff/Senior Rust Engineer with 11+ years building secure, high-performance platforms across application security, connected mobility, device protection, and SaaS using **Rust 1.x**, **Rust editions 2015–2021**, **Tokio**, **Axum**, **Actix Web**, **Kafka**, **PostgreSQL**, **AWS**, and **Kubernetes**, delivering memory-safe services, distributed systems, legacy modernization, performance optimization, production reliability, and technical leadership for business-critical products.

SKILLS

- **Programming Languages:** Rust, C++, Python, Go, SQL, Bash, TypeScript, Java
- **Rust Development:** Rust 1.x, Rust editions 2015–2021, ownership and borrowing, lifetimes, traits, generics, macros, Cargo workspaces, async/await, unsafe Rust review
- **Backend Frameworks:** Axum, Actix Web, Rocket, Warp, Tokio, Hyper, Tower, Tonic, Serde, SQLx, Diesel
- **APIs and Integration:** REST, gRPC, Protocol Buffers, GraphQL, WebSockets, OpenAPI, webhooks, API versioning, request validation, rate limiting
- **Distributed Systems:** Microservices, event-driven architecture, service-oriented architecture, domain-driven design, clean architecture, CQRS, multi-tenant SaaS
- **Messaging and Streaming:** Apache Kafka, RabbitMQ, AWS SQS, AWS SNS, consumer groups, event replay, schema evolution, idempotent consumers, dead-letter queues
- **Concurrency and Performance:** Tokio runtime, async I/O, multithreading, channels, lock-free structures, worker pools, backpressure, zero-copy processing, memory profiling
- **Databases and Caching:** PostgreSQL, MySQL, SQL Server, DynamoDB, Cassandra, Redis, Elasticsearch, query optimization, indexing, transactions
- **Cloud Platforms:** AWS EKS, ECS, EC2, Lambda, API Gateway, S3, RDS, DynamoDB, ElastiCache, IAM, CloudWatch, Azure Kubernetes Service
- **DevOps and Infrastructure:** Docker, Kubernetes, Helm, Terraform, GitHub Actions, GitLab CI/CD, Jenkins, Argo CD, Linux, Nginx
- **Testing and Quality:** Cargo test, Criterion, Proptest, Mockall, integration testing, contract testing, property-based testing, fuzz testing, Clippy, rustfmt
- **Observability:** OpenTelemetry, Prometheus, Grafana, Datadog, CloudWatch, ELK Stack, distributed tracing, structured logging, SLI/SLO management
- **Security:** OAuth 2.0, OpenID Connect, JWT, RBAC, TLS, secrets management, OWASP, dependency scanning, secure coding, supply-chain security
- **Leadership Practices:** Architecture reviews, technical roadmaps, ADRs, code reviews, production incident response, mentoring, Agile delivery, stakeholder collaboration

EXPERIENCE

Veracode Systems Inc — Senior Software Engineer

Remote/ Full time

April 2021 – May 2026

Somerville, MA

- Architected memory-safe application-security services using **Rust 2021**, **Tokio**, **Axum**, **Tonic**, and **PostgreSQL** for scan orchestration, findings enrichment, policy evaluation, and secure enterprise integrations.
- Migrated performance-sensitive components from legacy Java and Python services into independently deployable Rust microservices while preserving existing APIs and customer workflow compatibility.
- Reduced median scan-processing latency by **46%** through asynchronous I/O, bounded worker pools, zero-copy parsing, batch persistence, optimized serialization, and targeted Redis caching.

- Resolved a production memory-growth issue caused by retained task handles and unbounded message queues by introducing cancellation propagation, queue limits, heap analysis, and load-based regression tests.
- Designed event-driven processing with **Apache Kafka and AWS SQS**, supporting idempotent consumers, ordered workflows, retries, dead-letter handling, and replay-safe scan events.
- Built high-throughput **REST and gRPC APIs** with Axum, Tonic, Serde, Tower middleware, request validation, authorization controls, rate limiting, and structured error contracts.
- Improved PostgreSQL performance by replacing repeated queries with bulk operations, adding targeted indexes, reducing transaction contention, and separating reporting traffic from operational workloads.
- Deployed containerized services to **AWS EKS** using Docker, Helm, Terraform, GitHub Actions, autoscaling, readiness probes, canary releases, and automated rollback safeguards.
- Implemented observability with **OpenTelemetry, Prometheus, Grafana, and CloudWatch**, exposing service latency, queue saturation, error rates, memory use, and distributed request traces.
- Led architecture reviews, production incident investigations, Rust code-quality standards, and developer mentoring while coordinating delivery across security, product, QA, and SRE teams.

Cubic Telecom USA — Software Engineer

Remote/ Full time

January 2019 – April 2021

San Diego, CA

- Developed connected-vehicle and IoT services using **Rust 2018, Actix Web, Tokio, Kafka, and PostgreSQL** for device activation, subscription management, roaming, and telemetry workflows.
- Designed horizontally scalable ingestion services that validated and normalized high-volume modem, carrier, and network events before routing data to billing and analytics platforms.
- Increased event-processing throughput by **38%** through partition-aware consumers, asynchronous batching, compact serialization, reduced allocation, and carefully tuned concurrency limits.
- Eliminated duplicate carrier provisioning caused by retry storms and delayed acknowledgments by implementing idempotency keys, deduplication windows, and explicit state transitions.
- Built Kafka producers and consumers with schema compatibility, dead-letter handling, replay controls, backpressure, and operational metrics for globally distributed integration workflows.
- Developed secure REST APIs and partner adapters with Actix Web, Serde, OAuth 2.0, request signing, rate limiting, and detailed diagnostics for external mobile-network operators.
- Optimized PostgreSQL, Cassandra, and Redis access by selecting appropriate consistency models, reducing hot-key contention, improving indexes, and caching reference data.
- Containerized services with Docker and deployed them through Kubernetes pipelines on AWS, adding resource limits, health checks, rolling upgrades, and environment-specific configuration.
- Collaborated with product managers, SRE teams, telecom specialists, and QA engineers to estimate features, review designs, troubleshoot incidents, and coordinate global releases.

Asurion — Software Developer

Remote/ Full time

January 2017 – December 2018

Nashville, TN

- Developed backend services using **Rust 1.x, Rocket, Diesel, PostgreSQL, and Redis** for device-protection claims, eligibility validation, fulfillment, and customer-support workflows.
- Introduced Rust services alongside established Java and .NET applications for performance-sensitive validation and integration tasks without requiring a disruptive platform replacement.
- Reduced average claims-processing time by **26%** through parallel validation, cached reference data, optimized database access, and asynchronous execution of notification workflows.
- Resolved an intermittent double-fulfillment production defect caused by concurrent retries and non-atomic status updates by adding row locking, idempotent commands, and explicit workflow states.
- Built message-driven processing with RabbitMQ and Kafka for claims events, inventory updates, repair routing, and customer notifications, including failure isolation and retry controls.
- Improved service maintainability by separating transport, domain, and persistence logic while introducing typed errors, reusable traits, configuration validation, and structured logging.

- Expanded automated quality checks with unit tests, integration tests, property-based testing, Clippy, formatting enforcement, and CI pipelines for dependable releases.
- Partnered with analysts, support teams, and senior engineers to refine requirements, investigate production failures, review code, and document operational recovery procedures.

Rustici Software — Junior Software Developer

Hybrid/ Full time

June 2015 – December 2016

Nashville, TN

- Developed backend utilities with **early stable Rust, Python, PostgreSQL, and MySQL** for e-learning content import, course launching, activity tracking, and reporting workflows.
- Created command-line tools and lightweight services for validating SCORM packages, processing xAPI statements, and transforming large learning-content archives.
- Modernized selected parsing and batch-processing tasks from legacy scripts into compiled Rust components, improving type safety and reducing failures from malformed input data.
- Investigated content-import defects involving invalid XML, inconsistent encodings, oversized packages, missing metadata, and timezone conversion errors across customer environments.
- Improved processing performance by streaming large files, reducing unnecessary copies, batching database writes, and adding indexes for frequently accessed course metadata.
- Added unit tests, fixture-based parser coverage, reproducible Cargo builds, CI checks, and structured error reporting to improve release confidence and support diagnostics.
- Worked closely with senior developers and customer-support specialists to reproduce production issues, document root causes, review fixes, and deliver maintainable enhancements.

EDUCATION

Tennessee State University — Bachelor's Degree in Computer Science

Sep 2011 – May 2015

CERTIFICATION

- **AWS Certified Developer – Associate** — Amazon Web Services
- **Microsoft Certified: Azure Developer Associate** — Microsoft
- **Certified Kubernetes Application Developer — CKAD** — The Linux Foundation and Cloud Native Computing Foundation